# Constitutional Science

## A Governance Analysis of Popperian, Bayesian, and Kuhnian Method

**Each encodes a distinct refusal discipline and distributes its oracle dependencies differently.**

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## Abstract

Debates over the "scientific method" are often framed as philosophical disagreements about falsification, probability, or paradigm change. These disputes typically remain content-level arguments–what counts as evidence, how belief should update, or whether revolutions are rational. This paper proposes a structural reframing. When expressed in constitutional form, the Popperian, Bayesian, and Kuhnian formulations of scientific practice reveal themselves not as mutually exclusive doctrines, but as governance layers operating within a single epistemic system.

Each formulation locates authority differently: at the level of empirical integrity, probabilistic adjudication, or community standard evolution. Each encodes a distinct refusal discipline and distributes its oracle dependencies differently. By rendering these features structurally–through invariants, admissibility gates, oracle declarations, and outcome transitions–we expose complementary strengths and characteristic failure modes. The resulting insight is that scientific objectivity is not metaphysical purity but auditable dependency discipline. The scientific method, properly understood, is constitutional.

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## 1. Introduction: From Method to Constitutional Placement

The phrase "the scientific method" suggests a unified procedure. Yet historical and contemporary practice reveals multiple competing formulations:

The falsification-centered approach associated with Karl Popper.

The probabilistic updating framework grounded in Bayesian inference.

The sociotechnical paradigm analysis articulated by Thomas Kuhn.

These are frequently presented as rival epistemologies. The debate is cast as a dispute over the correct philosophical account of science.

However, once expressed structurally–defined by invariants, admissibility gates, oracle declarations, and refusal semantics–the dispute reframes itself. The three formulations do not primarily disagree about evidence. They relocate epistemic authority.

Popper governs at the level of empirical integrity. Bayes governs at the level of belief adjudication. Kuhn governs at the level of communal standards and framework evolution.

The question is not which is correct in isolation. The question is where each places constitutional power within the scientific enterprise.

This paper does not argue for a victor. It clarifies the jurisdictional structure.

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## 2. The Popperian Core: Integrity Through Refusal

The Popperian formulation privileges falsifiability. Structurally, it enforces:

Operationalizable claims.

Testable predictions under defined protocols.

Admissibility of disconfirming evidence.

Replication as status reinforcement.

Authority here is placed at the boundary of empirical integrity. A claim that cannot be tested cannot enter the jurisdiction of science.

The power of this structure lies in refusal discipline. It refuses unfalsifiable inputs. It insists that adjudication events be documented and replicable. Its backbone is constraint.

Yet this discipline introduces brittleness. Real-world models are approximate; confounds are complex; statistical noise blurs clean falsification. In practice, strict falsification often collapses into ritualized thresholding rather than decisive disproof.

Failure mode signature: rigidity under model ambiguity.

Popperian science locates constitutional authority in the integrity gate. It protects entry into the system.

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## 3. The Bayesian Core: Adjudication as Policy

The Bayesian formulation reframes scientific reasoning as belief update. Structurally, it requires:

Explicit priors.

Declared likelihood models.

Replayable posterior updates.

Policy-defined decision thresholds.

Authority here shifts to adjudication mechanics. Evidence does not destroy hypotheses; it reshapes probability mass.

A central structural insight emerges:

Decision thresholds are policy, not mathematics.

A posterior probability does not compel status transition. A declaration of "confirmed" or "disconfirmed" requires an explicit decision rule–often tied to loss functions, cost asymmetries, or institutional norms. What appears purely statistical is constitutionally governed.

The strength of Bayesian structure lies in continuity. It accommodates uncertainty and partial evidence. It prevents premature binary collapse.

Yet it introduces a new vulnerability: mathematical formalism can obscure hidden modeling assumptions. Priors may be smuggled under neutrality. Likelihood functions may encode unexamined preprocessing choices.

Failure mode signature: formal rigor masking undeclared dependencies.

Bayesian science locates constitutional authority in the update rule and decision policy.

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## 4. The Kuhnian Layer: Standards as Mutable Law

Kuhn's analysis shifts focus to the community level:

Paradigms define legitimate problems and exemplars.

Anomalies accumulate under normal science.

Crises destabilize confidence.

Revolutions restructure evaluative standards.

Authority here is placed at the level of communal norms and interpretive judgment.

Structurally, this introduces a decisive feature: standards drift. Evaluation criteria are not static invariants; they are versioned artifacts within communities.

The Kuhnian layer does not replace empirical adjudication. It governs the meta-conditions under which adjudication occurs. It describes how the constitution of science itself evolves.

However, without constraint, sociological dominance can impersonate epistemic authority. If community consensus overrides empirical integrity gates, constitutional collapse follows.

Failure mode signature: sociological authority substituting for evidentiary discipline.

Kuhn locates constitutional power at the level of standards governance.

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## 5. Oracle Surface Analysis

An oracle, in governance terms, is an authority whose outputs are accepted without internal proof.

Scientific methods do not eliminate oracles. They distribute them differently:

Popperian: oracles appear in operational definitions and measurement calibration.

Bayesian: oracles appear in prior selection, likelihood modeling, and threshold choice.

Kuhnian: oracles appear in standard-setting bodies and interpretive judgments about crisis.

Objectivity is therefore not the absence of oracles. It is oracle visibility.

When priors are undeclared, when standards drift silently, when threshold policies masquerade as inevitabilities, the oracle surface becomes hidden. Authority becomes unaccountable.

The constitutional question is not whether judgment exists. It is whether judgment is declared and contestable.

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## 6. Refusal Semantics as Constitutional Clause

Each formulation defines what the system must refuse:

Popper refuses unfalsifiable claims.

Bayesian refuses updates without declared priors and likelihoods.

Kuhnian refuses paradigm-neutral pretense.

Refusal semantics operate as constitutional clauses. They define admissibility boundaries.

When refusal collapses, dogma forms. When refusal overextends, inquiry ossifies.

Balanced science requires:

1. Operational discipline (entry integrity).
2. Transparent update policy (adjudication).
3. Versioned standard governance (meta-level adaptation).

These are not rival principles. They are coordinated constitutional safeguards.

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## 7. Replication Across Jurisdictions

Replication's meaning shifts with constitutional placement:

Under Popper, replication reinforces falsification resistance.

Under Bayes, replication reshapes posterior distributions.

Under Kuhn, replication patterns influence anomaly accumulation and paradigm stability.

The so-called "replication crisis" cannot be understood without specifying jurisdiction. A failure at one layer may signal instability in another.

Structural clarity prevents category error. It distinguishes empirical failure from adjudicative recalibration and from paradigm-level tension.

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## 8. Incommensurability as Structured Delta

Kuhn's concept of incommensurability is often interpreted as epistemic rupture. Structurally expressed, it becomes a mapping problem:

Terms partially fail to translate.

Standards diverge.

Evaluation criteria mismatch.

Rather than mystifying the divide, we can record it as structured delta. Incommensurability becomes traceable difference rather than incomprehensible rupture.

This transforms paradigm comparison from philosophical impasse into governance mapping.

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## 9. Science as Layered Constitutional Architecture

The comparative analysis reveals a composite constitutional structure:

Layer 1 – Empirical Integrity Operationalization, provenance, replication discipline.

Layer 2 – Adjudicative Update Belief revision under declared probabilistic or threshold rules.

Layer 3 – Standards Governance Community-level drift, anomaly management, and framework revision.

Confusion arises when one layer claims supremacy:

Falsification alone cannot govern sociotechnical evolution. Probabilistic updating cannot define communal legitimacy. Sociological description cannot replace empirical integrity.

Scientific resilience emerges when these jurisdictions remain distinct yet coordinated.

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## 10. Objectivity as Auditable Humility

Objectivity is often imagined as detachment from human influence.

Structural comparison suggests a different formulation:

Objectivity is auditable humility.

It consists of:

Declared dependencies.

Receipted adjudications.

Transparent decision thresholds.

Explicit oracle references.

Structured refusal and defer pathways.

Just as constitutional democracies rely on recorded deliberation rather than divine right, scientific objectivity relies on documented process rather than metaphysical neutrality.

Objectivity is not the elimination of judgment. It is judgment constrained by declared procedure.

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## 11. Implications Beyond Science

When expressed constitutionally, epistemologies become comparable governance systems.

Legal systems can be examined for hidden oracles. Economic models can be evaluated for threshold governance. Political systems can be inspected for silent standard drift.

Disagreement at the level of belief often masks disagreement at the level of constitutional placement.

Clarifying jurisdiction clarifies conflict.

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## 12. Conclusion

The Popperian, Bayesian, and Kuhnian formulations of scientific method do not compete as alternative truths about science. They describe distinct governance regimes operating at different layers of epistemic practice.

From their structural comparison, several insights follow:

Science always contains policy decisions.

Oracles cannot be eliminated; they must be declared.

Replication, falsification, and probabilistic updating are complementary mechanisms.

Paradigm shifts are governed events, not mystical ruptures.

Objectivity is procedural accountability, not metaphysical neutrality.

Science is not a single method. It is a layered constitutional system of admissibility, adjudication, and standards evolution.

Its resilience does not arise from the absence of human judgment.

It arises from forcing judgment to declare its authority.

## Standing and Claim Boundary

This paper is a conceptual, formal, or architectural contribution unless it explicitly binds a statement to a cited external source or an exact carried evidence artifact. Unbound statements about implementation, validation, performance, safety, universality, or proof are research claims or design obligations, not established operational facts. Corpus-internal references establish conceptual lineage and comparison, not independent external confirmation.

## Realization Scope and Evidence Boundary

This successor makes the paper's stage boundary explicit. It narrows claims rather than manufacturing missing evidence: a stage named as context is not thereby claimed as implemented, validated, or complete.

- **Possibility Space.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.
- **Publication form.** This work is retained as a standalone technical companion because its concrete migration method is useful independently; it remains subordinate to the architecture it explains.

## Corpus-Wide Reference Positioning

This paper argues that debates over the "scientific method" are often framed as philosophical disagreements about falsification, probability, or paradigm change. Its nearest external comparison fields are governance, software, provenance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

### Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. Modularity, secure-development practice, and reproducibility reduce classes of failure but do not guarantee architectural coherence or user sovereignty. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

### Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant

work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

## References

### External scholarship, standards, and prior art


[**E1**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E2**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

[**E3**] Lawrence Lessig. *Code: Version 2.0*. Basic Books, 2006. https://codev2.cc/ **Relationship:** Predecessor argument that software architecture regulates behavior; the paper's contribution may be to make such regulation inspectable and revisable.

[**E4**] OECD. *Recommendation of the Council on Artificial Intelligence*. OECD/LEGAL/0449, adopted 2019 and updated 2024. legalinstruments.oecd.org source **Relationship:** Adjacent policy standard for accountability, traceability, transparency, safety, and human-centered values.

[**E5**] David L. Parnas. "On the Criteria To Be Used in Decomposing Systems into Modules." *Communications of the ACM* 15(12), 1972, pp. 1053-1058. DOI: 10.1145/361598.361623. https://doi.org/10.1145/361598.361623 **Relationship:** Foundational modularity treatment used to compare information hiding with explicit semantic and governance interfaces.

[**E6**] Ward Cunningham. "The WyCash Portfolio Management System." *OOPSLA '92 Experience Report*, 1992. DOI: 10.1145/157709.157715. https://doi.org/10.1145/157709.157715 **Relationship:** Original technical-debt metaphor used to distinguish implementation debt from semantic debt.

[**E7**] Karen Scarfone, Murugiah Souppaya, and Donna Dodson. *Secure Software Development Framework (SSDF) Version 1.1*. NIST SP 800-218, 2022. DOI: 10.6028/NIST.SP.800-218. https://doi.org/10.6028/NIST.SP.800-218 **Relationship:** Adjacent secure-development framework for preparation, protection, production, and vulnerability response.

**Ableman corpus**

[**A1**] Adam Ableman. *Ableman Constitutional Governance*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-79353AF163A4B0F1;
source SHA-256
FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B.
**Corpus relationship:** priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Theory*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-D2D97C4042FD281F;
source SHA-256
FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF.
**Corpus relationship:** priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Governance Engineering*. Ableman Corpus current edition; work identity PAPER.PUB.18749870;
source SHA-256
372169F315D192733856B1F4A3F1038BBC73F57CF3ED9ACABAE72B965A69C8D0.
**Corpus relationship:** priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Universal Governance Kernel (UGK)*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-1F9BFB2BF8370226;
source SHA-256
AA5106EA71E5DF120E141435C280DFC532715CA5DE0C75B8862BB7784ECCFACD.
**Corpus relationship:** priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Legitimacy Without Self-Ratification*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-EF2382F9397335AC;
source SHA-256
0186A39C808F8FDDB5C8F42384DF1A26678928434BD50A0EAE8D4947390F68A2.
**Corpus relationship:** priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Sovereign Software*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-24C6159B98023A9C;
source SHA-256
6AAFE79E438B980B75D1910A897F7BA716A10CF1187B68CB1F71405F36725A18.
**Corpus relationship:** priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Constitutional Software Analysis (CSoftA): A Governance-Based Classification Method for Software Systems.* Ableman Corpus current edition; work identity `PAPER.PUB.20472196`;
source SHA-256
`3670B28B7C1AF5467BE4002C47B5F964F8FAF6F257125C2F84CE06904E4FD0DE`.
**Corpus relationship:** priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

**Editorial source notes retained from predecessor**

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.


Adam Ableman *A Theory of Agentic Observation.* Zenodo, 2026. doi:10.5281/zenodo.20437813

Adam Ableman *CARA: Constrained Artificial Reasoning Architecture.* Zenodo, 2025. doi:10.5281/zenodo.18099715

Adam Ableman *Constitutional Computation.* Zenodo, 2026. doi:10.5281/zenodo.20624488

Adam Ableman *Constraint-Bearing Composability.* Zenodo, 2026. doi:10.5281/zenodo.21632061

Adam Ableman *Fractal Governance Architecture.* Zenodo, 2026. doi:10.5281/zenodo.20754189

Adam Ableman *Governed Semantic Compilation: Lawful Continuation, Refusal Semantics, and Epistemic Promotion in Large Language Models.* Zenodo, 2026. doi:10.5281/zenodo.18453681

Adam Ableman *MCIR: A Canonical Intermediate Representation for Governed Transformation.* Zenodo, 2026. doi:10.5281/zenodo.21095303

Adam Ableman *Semantic Coordinate Identity Tokenization (SCIT).* Zenodo, 2026. doi:10.5281/zenodo.20532070


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Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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DOI: 10.5281/zenodo.23848670